

The Department Name is the Department of Electronic Engineering under the Faculty of Information & Communication Technology (FICT), and the program is fully accredited by the Pakistan Engineering Council (PEC). The core technological focus includes Robotics, Artificial Intelligence, Digital Manufacturing, Automated and Electric Vehicles, Information Security, Wireless Networks, and Nano-Technologies. Established in 2003, the department focuses on meeting the shifting technological needs of industry and society, playing a primary role in technical education within Balochistan. Graduates are employed across diverse sectors, including national and international research institutions, telecom companies, banking, automotive, textile, biomedical, petrochemical, manufacturing, freelancing, and entrepreneurship. The department aims to produce electronic engineers who can achieve specific Program Educational Objectives, which include PEO-1 to contribute their knowledge for the betterment of humanity in general and society in particular, PEO-2 to impart their analytical and innovation skills for the enhancement of research, development, and production of Electronic Systems, and PEO-3 to work efficiently in teams or individually, integrating effectively in diverse cultures and environments.

For the Bachelor of Science (BS) program admission requirements, candidates need F.Sc Pre-Engineering completed from any recognized board or equivalent with a minimum of 60% marks, or a Diploma of Associate Engineer (DAE) completed in a relevant field with at least 60% marks. The degree requirements involve a total of 136 credit hours across 45 courses with a minimum CGPA of greater than or equal to 2.0. In the curriculum plan, credit hour listings follow the format of theory plus lab. The first semester includes HUM-163 / MATHP-118 Functional English / Calculus and Analytical Geometry with no prerequisites and 3+0 / 3+0 credit hours, EE-112 & EE-112L Linear Circuit Analysis with no prerequisite and 3+1 credit hours, PHY-205 & PHY-205L Applied Physics with no prerequisite and 2+1 credit hours, and CS-123 / ELE-102L Applications of ICT / Workshop Practice with no prerequisites and 2+1 / 0+1 credit hours. The second semester includes HUM-268 / HUM-101 Communication Skills / Islamic Studies with no prerequisites and 3+0 / 2+0 credit hours, EE-208 & EE-208L Electronic Devices & Circuits with no prerequisite and 3+1 credit hours, ENGG-217L / HUM-102 Engineering Drawing / Pakistan Studies with no prerequisites and 0+1 / 2+0 credit hours, and MATHP-111 Linear Algebra with no prerequisite and 3+0 credit hours. The third semester includes STAT-403 / MATHA-214 Probability & Random Variables / Differential Equations with no prerequisite for the former and Calculus, Geometry, & Linear Alg for the latter and 3+0 / 3+0 credit hours, EE-110 & EE-110L Digital Logic Design with no prerequisite and 3+1 credit hours, EE-204 & EE-204L Instrumentation & Measurements with Electronic Devices & Circuits as a prerequisite and 2+1 credit hours, and EE-109 & EE-109L Electronic Circuit Design with no prerequisite and 3+1 credit hours.

The fourth semester includes HUM-276 / MATHA-213 TRW & PS / Complex Variables and Transform with no prerequisites and 3+0 / 3+0 credit hours, EE-491 & EE-491L Integrated Electronics Analog with Electronic Circuit Design as a prerequisite and 2+1 credit hours, EE-201 & EE-201L Electrical Network Analysis with Linear Circuit Analysis as a prerequisite and 3+1 credit hours, and CE-324 & CE-324L Microprocessors and Microcontrollers with Digital Logic Design as a prerequisite and 3+1 credit hours. The fifth semester includes EE-324 & EE-324L Electrical Machines with no prerequisite and 3+1 credit hours, STAT-403 Signal & Systems with no prerequisite and 2+1 credit hours, EE-241 & EE-241L Control Systems with Linear

Algebra as a prerequisite and 3+1 credit hours, CS-212 & CS-212L Object Oriented Programming for Engineers with no prerequisite and 3+1 credit hours, and TE-301 & TE-301L Data Communication & Networks with Communication Systems as a prerequisite and 2+1 credit hours. The sixth semester includes EE-421 Power Electronics with Electrical Machines as a prerequisite and 3+1 credit hours, MATHA-216 & L / EE-XX Numerical Analysis & Lab / IDEE-I with no prerequisite and 2+1 / 3+0 credit hours, TE-314 & TE-314L Communication Systems with Signals and Systems as a prerequisite and 3+1 credit hours, and MGMT-XX Management Elective-I with no prerequisite and 3+0 credit hours.

The seventh semester includes EE-XX / EE-XX Elective-I / Elective-II with no prerequisites and 3+0 / 3+0 credit hours, MGMT-XX Management Elective-II with no prerequisite and 3+0 credit hours, EE-441 & EE441L Industrial Electronics with no prerequisite and 3+1 credit hours, and XX-XX / EE-481L Social Science Elective-I / Final Year Project Part-I with no prerequisites and 2+0 / 0+3 credit hours. The eighth semester includes EE-XX / EE-XX Elective-III / Elective-IV with no prerequisites and 3+0 / 3+0 credit hours, TE-412 & TE-412L Digital Signal Processing with Signals & Systems as a prerequisite and 2+1 credit hours, XX-XX / XX-XX Social Science Elective-II / IDEE-II with no prerequisites and 2+0 / 3+0 credit hours, and EE-482L Final Year Project Part-II with Final Year Project Part-I as a prerequisite and 0+3 credit hours. For the Master of Science (MS) program admission requirements, the academic background must be a BE/BS degree in Electronic, Electrical, Computer, or Telecommunication Engineering from an HEC-recognized institution, academic performance requires a minimum of 60% marks under the annual grading system or a minimum CGPA of 2.5 out of 4.0, and standardized testing requires a valid GAT General score with at least a 50% cumulative result. MS degree requirements include a total of 30 credit hours, total courses of 8, and a minimum academic standing of CGPA greater than or equal to 2.5.